# Artificial Intelligence Software, Project 1

## Fine-tuning YOLO models on custom datasets

MIPT
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# Short Description

The goal of this assignment is fine-tuning a pretrained **Ultralytics YOLO model** on a custom dataset for a computer vision task of your choice. It's important to implement a **methodologically correct ML workflow**. The workflow must include:

- dataset collection
- dataset annotation
- dataset versioning
- experiment tracking
- model training and improvement
- deployment and demonstration

You must use:

- **Ultralytics YOLO**
- **ClearML** for dataset versioning and experiment tracking

# 1. Task Selection

Choose a computer vision task that can be demonstrated using:

- a **phone camera**, or
- a **webcam on your computer**

Students choose the task themselves.

Possible task types:

- object detection
- instance segmentation
- image classification

Important requirement:

The object or class must be **physically accessible**, since you will collect and annotate your own data.

# 2. Dataset Preparation

You must use both:

- an **open dataset**
- a **self-collected dataset**

Possible sources of open datasets:

- Kaggle
- Roboflow Universe
- Hugging Face Datasets
- other public datasets

Your own dataset must contain:

- **100–300 images minimum**
- different lighting conditions
- different backgrounds
- different object poses and viewpoints

# 3. Data Annotation and Versioning

All collected images must be **annotated**.

Possible annotation tools:

- CVAT
- Roboflow

Annotations must be exported in **Ultralytics YOLO format**.

Datasets must be **versioned using ClearML**.

You must create at least:

- one **initial dataset version**
- one **improved dataset version** (after additional data collection or re-annotation)

# 4. Baseline Model

Train a **baseline model** by fine-tuning a pretrained **Ultralytics YOLO model**.

Students may choose the model architecture depending on the task.

Examples:

- smaller models for faster inference
- larger models if computational resources allow

All training runs must be **logged in ClearML**.

# 5. Evaluation Metrics

Use the standard Ultralytics evaluation metrics

- Precision
- Recall
- mAP
- F1 score
- Confusion Matrix

Metrics must be tracked through **ClearML experiments**.

# 6. Experiments and Improvements

You must run at least **three experiments** aimed at improving the model.

Possible approaches:

- hyperparameter tuning
- data augmentation
- adding more training data
- fixing annotation errors
- balancing classes
- trying different YOLO model sizes

Each experiment must be:

- logged in **ClearML**
- reproducible

# 7. Training Resources

This assignment **can be completed on a CPU-only computer**.

Using additional resources is allowed but not required:

- GPU
- Google Colab
- remote servers

Since you are performing **fine-tuning**, training on CPU is feasible.

# 8. Deployment and Demonstration

Export the trained model using **Ultralytics**.

Demonstrate the solution in real time using:

- a **webcam**, or
- a **phone camera**

The system must show **live inference** detecting the selected objects.

# Submission

You must submit a **report** describing the full workflow.

The report must include:

- description of the selected task
- links to **ClearML project and experiments**
- dataset sources and dataset versions
- description of experiments
- evaluation metrics
- description of improvements
- demonstration of the final system (video or live demo)
- discussion of **difficulties and limitations**

The goal is to document the **entire ML workflow**.

# Deserve “A” grade!

– Oleg Bulichev

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